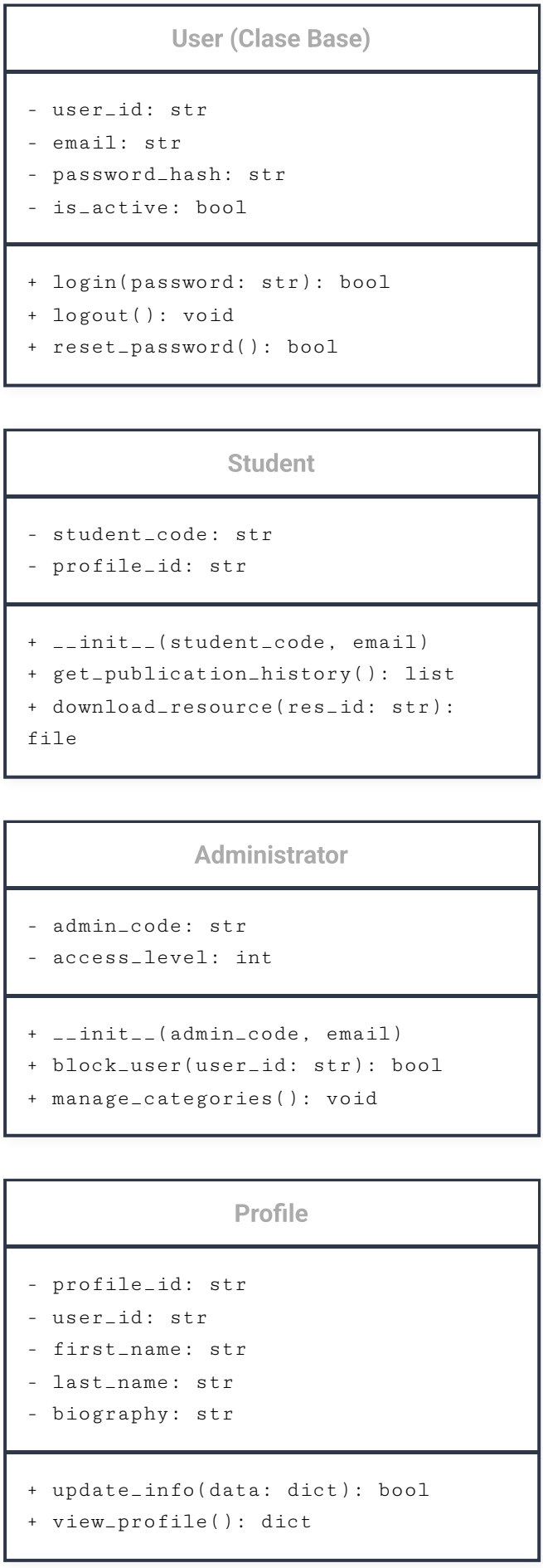
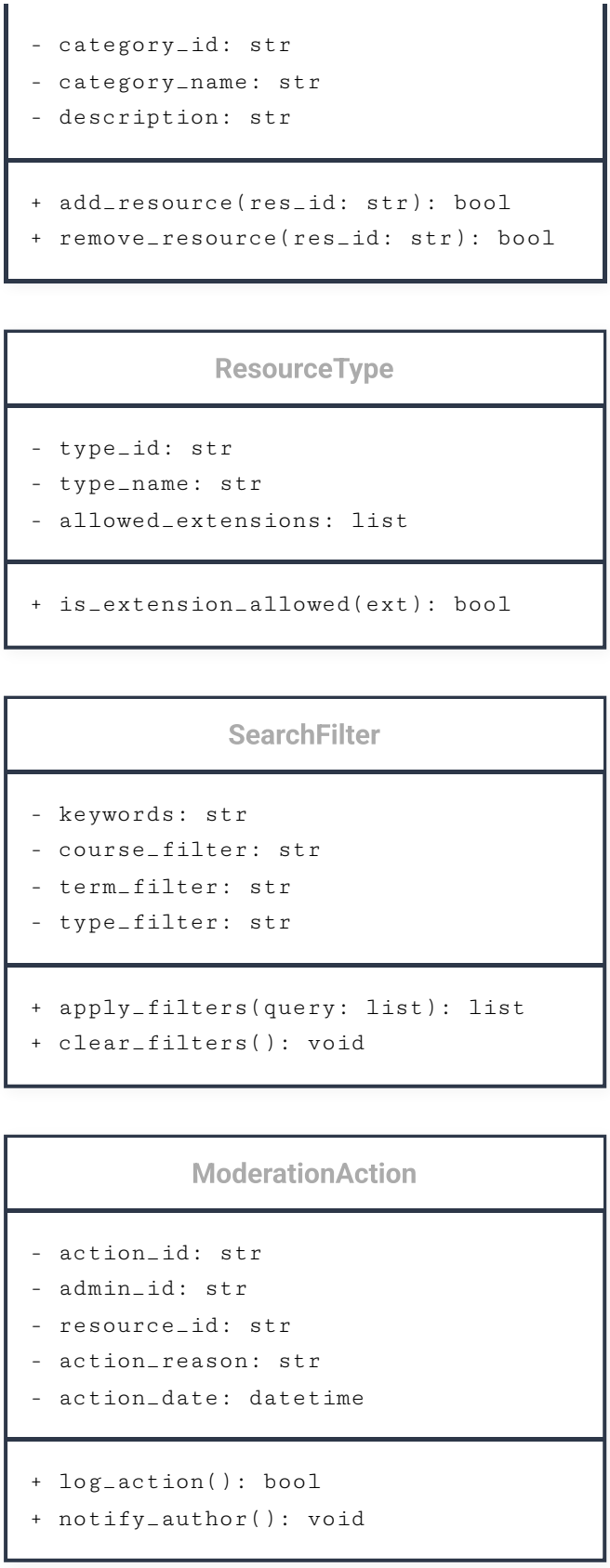

User (Clase Base)
- user_id: str
- email: str
- password_hash: str
- is_active: bool
+ login(password: str): bool
+ logout(): void
+ reset_password(): bool
Student
- student_code: str
- profile_id: str
+ __init__(student_code, email)
+ get_publication_history(): list
+ download_resource(res_id: str):
file
Administrator
- admin_code: str
- access_level: int
+ __init__(admin_code, email)
+ block_user(user_id: str): bool
+ manage_categories(): void
Profile
- profile_id: str
- user_id: str
- first_name: str
- last_name: str
- biography: str
+ update_info(data: dict): bool
+ view_profile(): dict


Resource
- resource_id: str
- title: str
- description: str
- author_id: str
- upload_date: datetime
- is_visible: bool
+ publish(): bool
+ edit_info(title, desc): bool
+ delete_resource(): bool
FileAttachment
- file_id: str
- resource_id: str
- file_path: str
- file_size_mb: float
- format_extension: str
+ validate_format(): bool
+ validate_size(): bool
+ get_download_link(): str
Course
- course_code: str
- course_name: str
- is_active: bool
+ get_resources(): list
+ update_course_info(): bool
AcademicTerm
- term_code: str
- start_date: date
- end_date: date
+ get_active_term(): str
+ get_resources_by_term(): list
Category


- category_id: str
- category_name: str
- description: str
+ add_resource(res_id: str): bool
+ remove_resource(res_id: str): bool
ResourceType
- type_id: str
- type_name: str
- allowed_extensions: list
+ is_extension_allowed(ext): bool
SearchFilter
- keywords: str
- course_filter: str
- term_filter: str
- type_filter: str
+ apply_filters(query: list): list
+ clear_filters(): void
ModerationAction
- action_id: str
- admin_id: str
- resource_id: str
- action_reason: str
- action_date: datetime
+ log_action(): bool
+ notify_author(): void